

B.Tech. Degree VII Semester Special Supplementary Examination April 2016

ME 1701 REFRIGERATION AND AIR CONDITIONING (2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) Explain the unit of refrigeration.
- (b) What do you understand by the principle of steam jet refrigeration?
- (c) List the differences between vapour compression and vapour absorption refrigeration systems.
- (d) Explain the types of refrigerant.
- (e) Differentiate the dry expansion and flooded evaporator with sketches.
- (f) What is a multi-stage compressor? Give its advantages.
- (g) Write a short note on by-pass factor for cooling coil.
- (h) What is effective temperature? What are the factors affecting effective temperature?

PART B

(4 × 15 = 60)

- II. Explain briefly the working of Bell Coleman air refrigerating cycle on p-v and T-S diagrams. Also derive an expression for COP. (15)

OR

- III. The capacity of a refrigerator is 450 tonnes when working between -15°C and 30°C . Find the mass of ice produced within 24 hours when water is supplied at 20°C . Also find the minimum power required. Assume the machine to be working on reversed Carnot cycle. Take latent heat of ice as 335 kJ/kg. (15)

- IV. In a standard vapour compression refrigeration cycle, operating between an evaporator temperature of -10°C and a condenser temperature of 40°C , the enthalpy of the refrigerant, Freon 12, at the end of compression is 220 kJ/kg. Draw the p-h and T-S diagrams. Calculate, (15)

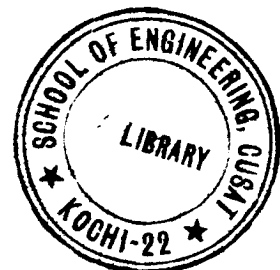
- (i) The COP of the cycle
- (ii) The refrigerating capacity and the compressor power assuming a refrigerant flow rate of 1 kg/min.

T($^{\circ}\text{C}$)	P(MPa)	h_f (kJ/kg)	h_g (kJ/kg)
-10	0.2191	26.85	183.1
40	0.9607	74.53	203.1

OR

- V. (a) Explain briefly with a neat diagram the working of practical aqua-ammonia vapour absorption system. (10)
- (b) Find the maximum COP of vapour absorption refrigeration in which heating, cooling and refrigeration takes place at the temperatures of 100°C , 20°C and -5°C respectively. (5)

(P.T.O.)



- VI. (a) Define volumetric efficiency of a reciprocating compressor and derive an expression for the same. (6)
- (b) Explain briefly the vane type compressor with diagram. (9)

OR

- VII. Explain briefly the types of condensers and expansion devices with sketches. (15)

- VIII. (a) Describe briefly the cooling and dehumidification process. (7)
- (b) The pressure and temperature of air in room are 1.0132 bar and 30°C respectively. If the relative humidity is found to be 40 percent, estimate; (8)
- (i) The partial pressure of water vapour.
 - (ii) Dew point temperature.
 - (iii) Specific volume of each component.
 - (iv) Specific humidity.

OR

- IX. Describe briefly the different methods of cooling load calculation. (15)